

eCargo: Delivering the future of Australasian logistics with end-to-end freight management

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ABOUT eCargo



eCargo executed seamless lift and migration, halved required for data processing, built proof disaster recovery system, and future Australasian expansion strategy with Google

About eCargo

eCargo, a division of Oji Fibre Solutions, is an enterprise software as a service (SaaS) business that builds freight management software to help manufacturers and retailers manage their freight. eCargo plays a critical part in New Zealand's supply chain. Some of their customers are New Zealand's most known brands like Foodstuffs, Countdown and Sanitarium, as well as manufacturing suppliers CHH Woodproducts and Oji Fibre Solutions.

Industries: Transportation

Google Cloud results

- Improved database processing performance after a lift and shift approach onto Compute Engine
- Improves system resilience with disaster recovery system that is 100% reliable under testing
- Inspires new expansion strategy with agile, granular Google Kubernetes Engine scaling

Halve requir datab proce tasks

Location: New Zealand

Storage (<https://cloud.google.com/storage/>),

dynamó

About Dynamo6

Dynamo6 is a New Zealand based tech company and Google Cloud partner. They provide services to empower organizations to innovate in the digital world. D6 services include tech strategy, cloud engineering, modern workplace, managed support, software development and digital experience.

ABOUT DYNAMO6

Google Cloud (<https://cloud.google.com/>)

Dynamo6 is a New Zealand based tech

Compute Engine

(<https://cloud.google.com/compute/>)

provide services to empower

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

cloud engineering, modern workplace,

Virtual Private Cloud

(<https://cloud.google.com/vpc/>)

"When switching to a cloud provider, we simply couldn't afford to get things wrong. So we did thorough performance tests, calculating reliability, agile scaling, cost-efficiency, intuitive use, availability and uptime. Of all major vendors, Google Cloud came out tops in each category. It was a straightforward decision for us."

—Tim Strawbridge,
Product Manager of SaaS
Enterprise Business,
eCargo

Logistics is the lifeblood of the consumer goods industry. From warehousing to transportation, businesses across industries rely on trusted freight solutions to keep them running. In fact, Australia's freight task is expected to grow by 25% between 2018 and 2040.

New Zealand-based eCargo

(<https://ecargo.co.nz/Info/customer-overviews/>) is on a mission to drive intelligent freight management solutions that enable global brands to execute more effectively. eCargo's freight management platform deploys proprietary algorithms to automate transport bookings, and bring control over freight costs.

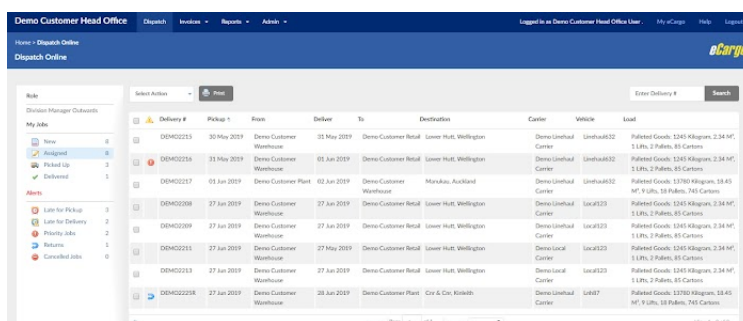
The eCargo platform acts as a middleman between the supplier's enterprise resource planning (ERP) systems and the transport management systems (TMS) of a network of freight suppliers . It can match a customer's orders with the best trucking solution to execute delivery, in the most timely and cost-effective manner possible. It's an end-to-end solution that gives suppliers and carriers full visibility of the freight lifecycle from pickup to delivery.

In 2021, eCargo learned that the data center that hosted its platform was shutting down. Faced with a scramble to find a new provider, eCargo decided that a move to the public cloud made sense, due to a

long experience of friction with virtual machine (VM) provisioning on physical servers. The need to manually configure VMs hampered the resilience and agility of the platform, which required zero downtime and seamless scalability to meet any fluctuation in freight traffic.

eCargo embarked upon a thorough head-to-head testing of all major cloud vendors. According to Tim Strawbridge, Product Manager of SaaS Enterprise Business, eCargo, it tapped Google Cloud (<https://cloud.google.com/>) based on a consideration of performance in key criteria, including resilience and ease-of-use. Working with partner Dynamo6 (<https://www.dynamo6.com/>), the decision brought major improvements in platform reliability, scalability, and disaster recovery systems.

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The screenshot displays the eCargo Dispatch Online interface. At the top, there's a navigation bar with 'Dispatch', 'Inventory', 'Reports', and 'Admin'. Below this, a sidebar on the left contains 'My Jobs' with filters for 'New', 'Assigned', 'Picked Up', and 'Delivered'. The main area shows a table of delivery jobs with columns for Delivery #, Pickup, From, Deliver, To, Destination, Carrier, Vehicle, and Load. The table lists several jobs with their respective dates, locations, and details.

Delivery #	Pickup	From	Deliver	To	Destination	Carrier	Vehicle	Load
DEM02253	30 May 2019	Demo Customer Warehouse	31 May 2019	Demo Customer Retail	Lower Hutt, Wellington	Demo Localhub Center	UnitHub432	Palletized Goods, 1245 Kilograms, 2.34 M ³ , 1 Lifts, 2 Pallets, 85 Cartons
DEM02254	31 May 2019	Demo Customer Warehouse	01 Jun 2019	Demo Customer Retail	Lower Hutt, Wellington	Demo Localhub Center	UnitHub432	Palletized Goods, 1245 Kilograms, 2.34 M ³ , 1 Lifts, 2 Pallets, 85 Cartons
DEM02257	01 Jun 2019	Demo Customer Plant	02 Jun 2019	Demo Customer Warehouse	Manukau, Auckland	Demo Localhub Center	UnitHub432	Palletized Goods, 15780 Kilograms, 18.45 M ³ , 9 Lifts, 18 Pallets, 740 Cartons
DEM02258	27 Jun 2019	Demo Customer Warehouse	27 Jun 2019	Demo Customer Retail	Lower Hutt, Wellington	Demo Localhub Center	UnitHub432	Palletized Goods, 1245 Kilograms, 2.34 M ³ , 1 Lifts, 2 Pallets, 85 Cartons
DEM02259	27 Jun 2019	Demo Customer Warehouse	27 Jun 2019	Demo Customer Retail	Lower Hutt, Wellington	Demo Localhub Center	UnitHub432	Palletized Goods, 1245 Kilograms, 2.34 M ³ , 1 Lifts, 2 Pallets, 85 Cartons
DEM02261	27 Jun 2019	Demo Customer Warehouse	27 Jun 2019	Demo Customer Retail	Lower Hutt, Wellington	Demo Localhub Center	UnitHub432	Palletized Goods, 1245 Kilograms, 2.34 M ³ , 1 Lifts, 2 Pallets, 85 Cartons
DEM02263	27 Jun 2019	Demo Customer Warehouse	27 Jun 2019	Demo Customer Retail	Lower Hutt, Wellington	Demo Localhub Center	UnitHub432	Palletized Goods, 1245 Kilograms, 2.34 M ³ , 1 Lifts, 2 Pallets, 85 Cartons
DEM02268	27 Jun 2019	Demo Customer Warehouse	28 Jun 2019	Demo Customer Plant	Civ & Co, Kaitiaki	Demo Localhub Center	UnitHub432	Palletized Goods, 1245 Kilograms, 2.34 M ³ , 1 Lifts, 2 Pallets, 85 Cartons
DEM02269	27 Jun 2019	Demo Customer Warehouse	28 Jun 2019	Demo Customer Plant	Civ & Co, Kaitiaki	Demo Localhub Center	UnitHub432	Palletized Goods, 1245 Kilograms, 2.34 M ³ , 1 Lifts, 2 Pallets, 85 Cartons

Achieving superior VM performance to drive the future of Australasian logistics

Initially, eCargo thought it had a natural choice among the major cloud providers because it already used the database management solution of a Google Cloud competitor. However, when the team carried out VM stress and performance tests on the rival service, it found that the results fell short of what was needed to entrust the future of a fast-growing logistics platform. When it carried out the same VM tests on Google Compute Engine (<https://cloud.google.com/compute>), Strawbridge said, eCargo met a "breath of fresh air."

"When we tried Compute Engine, we found that everything worked very well. The automation was straight forward and the performance was noticeably better. There was a great console to boot," says Strawbridge. "It did everything we expected and more."

The results achieved from the migration have been significant. From occasional system outages on the legacy data center, eCargo has achieved zero downtime even during the highest spikes in customer demand. Meanwhile, database performance has, in some cases, doubled thanks to Compute Engine's scalable computing power, with heavy data requests taking half as long as they previously did, says Strawbridge.

In addition to Compute Engine, the business deploys Pub/Sub (<https://cloud.google.com/pubsub>), Cloud

Storage (<https://cloud.google.com/storage>), and Virtual Private Cloud (<https://cloud.google.com/vpc>) for security firewalls, all of which play a key supporting role in eCargo's architecture modernization. Process automation is further boosted by Google Cloud compatibility of Terraform, a third-party infrastructure as code (IaC) solution that enables provisioning computing resources through machine-readable definition files, rather than physical hardware configuration.

"By using Terraform on Compute Engine, we were able to script the creation of all the VMs, and all the networking requirements and security firewalls, in automated fashion, with zero downtime during a complex infrastructure migration," says Strawbridge.

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Enabling robust disaster recovery for ultimate system resilience

One of the key areas where eCargo felt it needed a reliable and versatile cloud solution was disaster recovery (DR), an identical backup system that can kick into action in case of platform failure. In the legacy data center, eCargo felt vulnerable because tests on the provider's DR solutions often failed. Since DR was not fully automated, eCargo was at risk of human error creeping into the system.

Those concerns disappeared with the migration to Google Cloud, with its built-in disaster recovery protocols underpinned by a world class network of data centers. Partner Dynamo6 further enhanced system security, helping to architect a tailor-made DR strategy to create an identical recovery solution that works 100% of the time under testing.

"Trustworthy disaster recovery is one of the most critical needs in our platform, and it would often fail under testing on our legacy provider," says Bart Joy, Senior Engineer, eCargo. "Dynamo6 was instrumental in helping us build a robust DR environment that we can regularly test and have confidence in. Now we have peace of mind that we will be able to recover should we have a data center failure."

Enabling new Australasian growth paths with open-ended Google Cloud partnership

The migration to Compute Engine as eCargo's infrastructure was the first phase of an ongoing adoption of Google Cloud solutions. The next steps will be moving off a relational database management solution offered by a rival cloud provider to a Cloud SQL managed relational database on Google Cloud.

For years, Strawbridge says the DevOps team spent too much time working on platform updates and patches, distracting the team from the core mission of building innovative solutions for customers. On Cloud SQL, Strawbridge and his team are looking forward to seamless automated workflows and upgrades that will significantly improve operational efficiencies.

To further enhance eCargo's flexibility for agile scaling of the platform as it expands, the business is also planning to gradually move workloads to a containerized environment, powered by Google

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE). In addition to more seamless automated scalability, that is expected to enable it to optimize cost structures for clients thanks to more granular computing capabilities on GKE.

"We've already seen so many performance improvements both in terms of system resilience and database speed in just phase one of our move to Google Cloud," says Strawbridge. "We can't wait to see the exciting new paths we'll take in the next stages of this partnership, discovering the benefits of GKE and Cloud SQL."

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